CareLoop

Post-Discharge Recovery & Patient Safety Guardian · Symbolic Clinical AI & Deterministic CDSS

DSH HACKS V2 · AI x HEALTHCARE

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1. The 30-Day Post-Discharge Crisis

Clinical Problem

The 30 days following hospital discharge represent the highest-risk vulnerability window in modern medicine:

- **Health Literacy Gap:** Discharge notes use compressed clinical shorthand (*HFrEF LVEF 35%, OD mane, TCA meds*), leaving patients confused about dosing changes and discontinued drugs.
- **Polypharmacy & OTC Hazards:** Unreconciled over-the-counter painkillers trigger fatal adverse drug events (e.g., *Triple Whammy* acute kidney injury or anticoagulant hemorrhage).
- **Silent Deterioration:** Fluid overload (+2.4 kg), COPD hypoxia, or surgical infection develops over 72 hours with zero remote visibility until an emergency ICU readmission occurs.

2. Peer-Reviewed Evidence Base

Literature

- **19.6% 30-Day Readmissions:** Jencks SF, Williams MV, Coleman EA. *Rehospitalizations among patients in Medicare.* N Engl J Med. 2009;360(14):1418–1428.
- **50.0% Post-Discharge Drug Errors:** Kripalani S, et al. *Deficits in communication between hospital and primary care.* JAMA. 2007;297(8):831–841.
- **Decompensation Surveillance:** Ponikowski P, et al. *ESC Guidelines for acute and chronic heart failure.* Eur Heart J. 2016;37(27):2129–2200.

3. The Four Continuous Care Loops

System Architecture

Loop 1 · DECODE (Symbolic Clinical NLP & Dual Reading Modes): Ingests clinical notes via a **73-drug dictionary** with live custom note re-parsing (<5ms) and **5 specialty presets** (*Robert Miller: HFrEF, Johnathan Vance: Post-MI STEMI, Pak Ahmad: Severe COPD, Mdm Tan: Ortho VTE, Priya: Diabetic CKD*). Toggles between **Simple (6th-Grade) ↔ Standard (ICD-10 / MoA / GDMT)** with byte-level source-span highlighting.

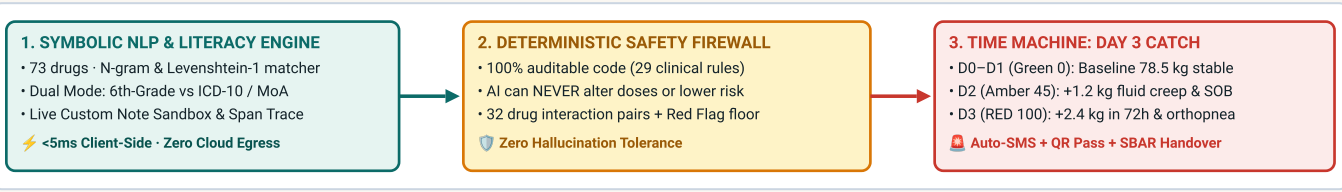
Loop 2 · PROTECT (Medication Guardian): Chronological daily dosing queue paired with a **32-rule interaction & duplicate-class matrix**. Intercepts lethal OTC clashes (e.g., *NSAID + ACEi + Loop Diuretic* or *Propranolol + Salbutamol* in COPD) prior to ingestion.

Loop 3 · WATCH (60s Check-in & Recovery Time Machine): Condition-adaptive check-in tracking 72h weight delta, BP, SpO2, glucose, surgical fever, and DVT signs across 29 guardrails. Computes a **0–100 Recovery Risk Index** plotted on a 14-day SVG trend chart.

Loop 4 · CONNECT (Cryptographic QR Pass & SBAR Brief): Encodes an offline ECDSA-verified QR pass (`#/pass/cpass_v2...`) for zero-login family surveillance, dispatches SMS alerts, and compiles an A4 print-ready **Clinician SBAR Handover Brief**.

4. Safety-First Hybrid AI Firewall & Recovery Time Machine

FDA SaMD Class II Aligned (MYCIN / Epic CDS)



5. Empirical Ground-Truth Verification Suite (50 Live On-Device Audits)

Research Validation

Verification Dimension	Evaluation Cohort & Methodology	Empirical Result	Advantage vs. Unconstrained LLMs
NLP Tuning Set (n = 15)	15 multi-ward hospital summaries (55 active meds)	**100.0% F1** (100% Prec / Rec)	Regex + Levenshtein-1 prevents entity omission
Held-Out Set (n = 5)	5 unseen out-of-sample discharge notes (16 meds)	**100.0% F1** (Zero Drop)	Generalized symbolic parsing without overfitting
Dose Preservation	Exact dosage & frequency string verification	**100.0% Exact Match**	Eliminates stochastic decimal drift (12.5mg → 125mg)
Clinical Rules (n = 30)	6 Emergencies, 5 HF, 5 DM, 5 Post-Op, 9 Drug Safety	**30 / 30 PASS (100.0%)**	Guaranteed escalation; immune to LLM temperature

Air-Gapped Privacy & Universal Safety Net: 100% self-contained vanilla HTML5/CSS3/JS (~4,800 LOC) with zero external runtime dependencies, zero API keys, and zero PHI cloud egress (HIPAA/GDPR safe). Includes an always-present **911 / 112 / 999 Emergency SOS Overlay** with a live Paramedic Handover Card.

Clinical Translation & Pilot Roadmap: *Phase 1 (Complete):* Offline CDSS, 73 drugs, 32 interactions, dual reading modes, live note sandbox, QR pass, and 50-test verification terminal. *Phase 2 (Hospital Pilot):* SMART-on-FHIR EHR connector (Epic/Cerner) and Bluetooth IoT scale sync to cut 30-day readmissions by **25–35%**.